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-- SECTION C : Aggregation Queries

-- Database : Celebal_Week2

-- Table : dbo.Superstore

-- Tool : SQL Server Management Studio (SSMS)

-- Author : ARATI THORAT | Celebal Internship – Week

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USE Celebal_Week2;

GO

-- _____

-- C1. Overall Business Summary

-- PURPOSE : Get a single-row KPI snapshot of the entire store

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SELECT

COUNT(*) AS Total_Transactions,
COUNT(DISTINCT Order_ID) AS Total_Orders,
COUNT(DISTINCT Customer_ID) AS Total_Customers,
COUNT(DISTINCT Product_ID) AS Total_Products,
ROUND(SUM(Sales), 2) AS Total_Revenue,
ROUND(SUM(Profit), 2) AS Total_Profit,
SUM(Quantity) AS Total_Units_Sold,
ROUND(AVG(Sales), 2) AS Avg_Sale_Per_Transaction,
ROUND(AVG(Profit), 2) AS Avg_Profit_Per_Transaction,
ROUND(MIN(Sales), 2) AS Min_Sale,
ROUND(MAX(Sales), 2) AS Max_Sale

FROM dbo.Superstore;

100 % 🟢 No issues found Ln: 1

	Total_Transactions	Total_Orders	Total_Customers	Total_Products	Total_Revenue	Total_Profit	Total_Units_Sold	Avg_Sale_Per_Transaction	Avg_Profit_Per_Transaction	Min_Sale	Max_Sale
1	9994	5009	793	1862	2297200.86	286817.02	37873	229.86	28.7	0.44	22638.48

OBSERVATION:

Total Revenue ≈ \$2.30 Million

Total Profit ≈ \$286,000

Overall profit margin ≈ 12.5%

9,994 transactions across 5,009 unique orders

Average sale per transaction ≈ \$230

-- C2. Sales and Profit by Category

-- PURPOSE : Compare revenue and profitability by product line

SELECT

Category,

COUNT(*) AS Total_Transactions,

SUM(Quantity) AS Units_Sold,

ROUND(SUM(Sales), 2) AS Total_Sales,

ROUND(SUM(Profit), 2) AS Total_Profit,

ROUND(AVG(Sales), 2) AS Avg_Sale,

ROUND(AVG(Profit), 2) AS Avg_Profit,

ROUND(MAX(Sales), 2) AS Max_Sale,

ROUND(MIN(Sales), 2) AS Min_Sale

FROM dbo.Superstore

GROUP BY Category

ORDER BY Total_Sales DESC;

Results		Messages								
	Category	Total_Transactions	Units_Sold	Total_Sales	Total_Profit	Avg_Sale	Avg_Profit	Max_Sale	Min_Sale	
1	Technology	1847	6939	836154.03	145454.95	452.71	78.75	22638.48	0.99	
2	Furniture	2121	8028	741999.79	18871.27	349.83	8.9	4416.17	1.89	
3	Office Supplies	6026	22906	719047.03	122490.8	119.32	20.33	9892.74	0.44	

OBSERVATION:

1. Technology → Highest revenue AND highest profit
2. Furniture → High revenue but LOW profit (losses on Tables)
3. Office Supp. → Most transactions, moderate profit

KEY INSIGHT: Furniture's high revenue doesn't translate to profit
due to excessive discounting on Tables and Bookcases.

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-- C3. Sales and Profit by Sub-Category

-- PURPOSE : Identify best and worst performing sub-categories

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SELECT

Category,

Sub_Category,

COUNT(*) AS Transactions,

ROUND(SUM(Sales), 2) AS Total_Sales,

ROUND(SUM(Profit), 2) AS Total_Profit,

ROUND(AVG(Discount), 4) AS Avg_Discount,

ROUND(SUM(Profit) * 100.0

/ NULLIF(SUM(Sales),0), 2) AS Profit_Margin_Pct

FROM dbo.Superstore

GROUP BY Category, Sub_Category

ORDER BY Total_Profit DESC;

100 % No issues found

Results Messages

	Category	Sub_Category	Transactions	Total_Sales	Total_Profit	Avg_Discount	Profit_Margin_Pct
1	Technology	Copiers	68	149528.03	55617.82	0.1618	37.2
2	Technology	Phones	889	330007.05	44515.73	0.1546	13.49
3	Technology	Accessories	775	167380.32	41936.64	0.0785	25.05
4	Office Supplies	Paper	1370	78479.21	34053.57	0.0749	43.39
5	Office Supplies	Binders	1523	203412.73	30221.76	0.3723	14.86
6	Furniture	Chairs	617	328449.1	26590.17	0.1702	8.1
7	Office Supplies	Storage	846	223843.61	21278.83	0.0747	9.51
8	Office Supplies	Appliances	466	107532.16	18138.01	0.1665	16.87
9	Furniture	Furnishings	957	91705.16	13059.14	0.1383	14.24
10	Office Supplies	Envelopes	254	16476.4	6964.18	0.0803	42.27
11	Office Supplies	Art	796	27118.79	6527.79	0.0749	24.07
12	Office Supplies	Labels	364	12486.31	5546.25	0.0687	44.42
13	Technology	Machines	115	189238.63	3384.76	0.3061	1.79
14	Office Supplies	Fasteners	217	3024.28	949.52	0.082	31.4
15	Office Supplies	Supplies	190	46673.54	-1189.1	0.0768	-2.55
16	Furniture	Bookcases	228	114880	-3472.56	0.2111	-3.02
17	Furniture	Tables	319	206965.53	-17305.48	0.2613	-8.36

OBSERVATION:

TOP PROFIT: Copiers, Phones, Accessories, Paper

LOSS-MAKING: Tables (-\$17,725), Bookcases (-\$3,473), Supplies (-\$1,189)

Tables have an average discount > 40% → direct cause of losses.

Copiers have the highest profit margin at ~37%.

-- C4. Sales and Profit by Region

-- PURPOSE : Identify strongest and weakest geographic markets--

```

SELECT

    Region,

    COUNT(DISTINCT Order_ID)  AS Total_Orders,

    COUNT(DISTINCT Customer_ID) AS Total_Customers,

    ROUND(SUM(Sales), 2)      AS Total_Sales,

    ROUND(SUM(Profit), 2)     AS Total_Profit,

    ROUND(AVG(Sales), 2)      AS Avg_Order_Sale,

    ROUND(SUM(Profit) * 100.0

        / NULLIF(SUM(Sales),0), 2) AS Profit_Margin_Pct

FROM  dbo.Superstore

GROUP BY Region

ORDER BY Total_Sales DESC;

```

100 % No issues found

	Region	Total_Orders	Total_Customers	Total_Sales	Total_Profit	Avg_Order_Sale	Profit_Margin_Pct
1	West	1611	686	725457.82	108418.45	226.49	14.94
2	East	1401	674	678781.24	91522.78	238.34	13.48
3	Central	1175	629	501239.89	39706.36	215.77	7.92
4	South	822	512	391721.91	47169.43	241.8	12.04

OBSERVATION:

1. West → Highest sales (\$725K) and profit (\$108K)
2. East → Second in sales and profit
3. Central → Lowest profit despite moderate sales (margin only ~7%)
4. South → Lowest sales volume

Central region underperforms on profit — likely over-discounting.

-- **C5. Sales and Profit by Customer Segment**

-- PURPOSE : Understand which segment drives the business

SELECT

Segment,

COUNT(DISTINCT Customer_ID) AS Total_Customers,

COUNT(DISTINCT Order_ID) AS Total_Orders,

ROUND(SUM(Sales), 2) AS Total_Sales,

ROUND(SUM(Profit), 2) AS Total_Profit,

ROUND(AVG(Sales), 2) AS Avg_Sale_Per_Transaction,

ROUND(SUM(Profit) * 100.0

/ NULLIF(SUM(Sales),0), 2) AS Profit_Margin_Pct

FROM dbo.Superstore

GROUP BY Segment

ORDER BY Total_Sales DESC;

100 % No issues found

Results

Messages

	Segment	Total_Customers	Total_Orders	Total_Sales	Total_Profit	Avg_Sale_Per_Transaction	Profit_Margin_Pct
1	Consumer	409	2586	1161401.34	134119.21	223.73	11.55
2	Corporate	236	1514	706146.36	92399.13	233.82	13.08
3	Home Office	148	909	429653.15	60298.68	240.97	14.03

OBSERVATION:

Consumer → 52% of total revenue (largest segment)

Corporate → Higher average order value, better margins

Home Office → Fewest customers but decent profit margin

RECOMMENDATION: Focus retention efforts on Corporate segment

as they generate the best profit per order.

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-- **C6. Monthly Sales Trend (use case: monthly trends)**

-- PURPOSE : Identify seasonal patterns in sales

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SELECT

YEAR(Order_Date) AS Order_Year,
MONTH(Order_Date) AS Order_Month,
DATENAME(MONTH, Order_Date) AS Month_Name,
COUNT(DISTINCT Order_ID) AS Total_Orders,
ROUND(SUM(Sales), 2) AS Monthly_Sales,
ROUND(SUM(Profit), 2) AS Monthly_Profit,
ROUND(AVG(Sales), 2) AS Avg_Sale

FROM dbo.Superstore

GROUP BY

YEAR(Order_Date),
MONTH(Order_Date),
DATENAME(MONTH, Order_Date)

ORDER BY Order_Year, Order_Month;

100 % No issues found Ln: 166, Ch: 1 (541 chars, 14 lines) SPC CRLF UTF-8

Results Messages

	Order_Year	Order_Month	Month_Name	Total_Orders	Monthly_Sales	Monthly_Profit	Avg_Sale
1	2014	1	January	32	14236.89	2450.19	180.21
2	2014	2	February	28	4519.89	862.31	98.26
3	2014	3	March	71	55691.01	498.73	354.72
4	2014	4	April	66	28295.34	3488.84	209.6
5	2014	5	May	69	23648.29	2738.71	193.84
6	2014	6	June	66	34595.13	4976.52	256.26
7	2014	7	July	65	33946.39	-941.48	237.39
8	2014	8	August	72	27909.47	5318.1	182.41
9	2014	9	September	130	81777.35	8328.1	305.14
10	2014	10	October	78	31453.39	3448.26	197.82
11	2014	11	November	151	78628.72	9292.13	247.26
12	2014	12	December	141	69545.62	8983.57	250.16
13	2015	1	January	29	18174.08	-3281.01	313.35
14	2015	2	February	36	11951.41	2813.85	186.74
15	2015	3	March	79	38726.25	9732.1	280.63
16	2015	4	April	72	34195.21	4187.5	213.72
17	2015	5	May	74	30131.69	4667.87	206.38
18	2015	6	June	68	24797.29	3335.56	179.69
19	2015	7	July	66	28765.32	3288.65	205.47
20	2015	8	August	68	36898.33	5355.81	232.06
21	2015	9	September	140	64595.92	8209.16	220.46
22	2015	10	October	87	31404.92	2817.37	189.19
23	2015	11	November	158	75972.56	12474.79	234.48
24	2015	12	December	161	74919.52	8016.97	237.09
25	2016	1	January	48	18542.49	2824.82	208.34
26	2016	2	February	45	22978.82	5004.58	276.85
27	2016	3	March	86	51715.88	3611.97	317.28
28	2016	4	April	89	38750.04	2977.81	227.94
29	2016	5	May	108	56987.73	8662.15	253.28
30	2016	6	June	97	40344.53	4750.38	202.74
31	2016	7	July	96	39261.96	4432.88	195.33
32	2016	8	August	90	31115.37	2062.07	176.79
33	2016	9	September	192	73410.02	9328.66	202.23
34	2016	10	October	105	59687.74	16243.14	304.53
35	2016	11	November	183	79411.97	4011.41	214.63

Query executed successfully. localhost/SQLEXPRESS (17.0 ... ARATT\HomePC (90) Celebal_Week2 00:00:00 Row: 1, Col: 1 48 rows

OBSERVATION:

Clear seasonality pattern:

→ Q4 (Oct, Nov, Dec) is consistently the highest-sales period

→ Q1 (Jan, Feb) is the slowest period each year

→ November and December spike due to holiday purchasing

This is valuable for inventory and staffing planning.

-- C7. Yearly Sales Summary

-- PURPOSE : Track year-over-year business growth

SELECT

YEAR(Order_Date) AS Order_Year,

COUNT(DISTINCT Order_ID) AS Total_Orders,

COUNT(DISTINCT Customer_ID) AS Total_Customers,


```

ROUND(SUM(Sales), 2)    AS Annual_Sales,
ROUND(SUM(Profit), 2)   AS Annual_Profit,
ROUND(AVG(Sales), 2)    AS Avg_Sale
FROM dbo.Superstore
GROUP BY YEAR(Order_Date)
ORDER BY Order_Year;

```

100 %   No issues found

 Results

 Messages

	Order_Year	Total_Orders	Total_Customers	Annual_Sales	Annual_Profit	Avg_Sale
1	2014	969	595	484247.5	49543.97	242.97
2	2015	1038	573	470532.51	61618.6	223.85
3	2016	1315	638	609205.6	81795.17	235.49
4	2017	1687	693	733215.25	93859.27	221.38

OBSERVATION:

Sales grew consistently from 2014 to 2017.

2017 shows the highest revenue (~\$733K in available months).

Customer base also grew year-over-year, confirming business expansion.

-- C8. Top 10 Best-Selling Products (by Revenue)

-- PURPOSE : Identify hero products that drive revenue

```
SELECT TOP 10
```

```
Product_ID,
```

```
Product_Name,
```

```

Category,

Sub_Category,

COUNT(*)          AS Times_Ordered,

SUM(Quantity)       AS Total_Units_Sold,

ROUND(SUM(Sales), 2) AS Total_Revenue,

ROUND(SUM(Profit), 2) AS Total_Profit

FROM dbo.Superstore

GROUP BY Product_ID, Product_Name, Category, Sub_Category

ORDER BY Total_Revenue DESC;

```

	Product_ID	Product_Name	Category	Sub_Category	Times_Ordered	Total_Units_Sold	Total_Revenue	Total_Profit
1	TEC-CO-10004722	Canon imageCLASS 2200 Advanced Copier	Technology	Copiers	5	20	61599.82	25199.93
2	OFF-BI-10003527	Fellowes PB500 Electric Punch Plastic Comb Binding ...	Office Supplies	Binders	10	31	27453.38	7753.04
3	TEC-MA-10002412	Cisco TelePresence System EX90 Videoconferencing ...	Technology	Machines	1	6	22638.48	-1811.08
4	FUR-CH-10002024	HON 5400 Series Task Chairs for Big and Tall	Furniture	Chairs	8	39	21870.58	0
5	OFF-BI-10001359	GBC DocuBind TL300 Electric Binding System	Office Supplies	Binders	11	37	19823.48	2233.51
6	OFF-BI-10000545	GBC Ibimaster 500 Manual ProClick Binding System	Office Supplies	Binders	9	48	19024.5	760.98
7	TEC-CO-10001449	Hewlett Packard LaserJet 3310 Copier	Technology	Copiers	8	38	18839.69	6983.88
8	TEC-MA-10001127	HP Designjet T520 Inkjet Large Format Printer - 24" C...	Technology	Machines	3	12	18374.9	4094.98
9	OFF-BI-10004995	GBC DocuBind P400 Electric Binding System	Office Supplies	Binders	6	27	17965.07	-1878.17
10	OFF-SU-10000151	High Speed Automatic Electric Letter Opener	Office Supplies	Supplies	3	11	17030.31	-262

OBSERVATION:

Top revenue products are Copiers and Phones (Technology).

Canon imageCLASS Copier alone generates significant revenue.

These high-ticket items drive disproportionate profit.

-- C9. Top 10 Customers by Total Purchase Value

-- PURPOSE : Identify VIP customers for retention strategy

```

SELECT TOP 10

    Customer_ID,

    Customer_Name,

    Segment,

    Region,

    COUNT(DISTINCT Order_ID) AS Total_Orders,

    SUM(Quantity)          AS Total_Units_Bought,

    ROUND(SUM(Sales), 2)   AS Total_Spent,

    ROUND(SUM(Profit), 2)  AS Total_Profit_Generated

FROM  dbo.Superstore

GROUP BY Customer_ID, Customer_Name, Segment, Region

ORDER BY Total_Spent DESC;

```

Results		Messages						
	Customer_ID	Customer_Name	Segment	Region	Total_Orders	Total_Units_Bought	Total_Spent	Total_Profit_Generated
1	SM-20320	Sean Miller	Home Office	South	2	27	23669.2	-1787.04
2	TC-20980	Tamara Chand	Corporate	Central	2	24	18437.14	8745.06
3	RB-19360	Raymond Buch	Consumer	West	2	23	14345.28	6807.09
4	TA-21385	Tom Ashbrook	Home Office	East	2	22	13723.5	4599.21
5	AB-10105	Adrian Barton	Consumer	Central	5	48	12181.59	5362.61
6	BM-11140	Becky Martin	Consumer	Central	1	37	10539.9	-1878.79
7	HL-15040	Hunter Lopez	Consumer	East	2	5	10522.55	5045.86
8	BS-11365	Bill Shonely	Corporate	East	2	21	10022.29	2558.58
9	SC-20095	Sanjit Chand	Consumer	Central	1	17	9900.19	4668.69
10	GT-14710	Greg Tran	Consumer	East	5	42	9382.93	1749.11

OBSERVATION:

Top customers are mostly in Consumer and Corporate segments.

Some top-spending customers actually generate negative profit

due to excessive discounts — needs review.

-- C10. Ship Mode Analysis

-- PURPOSE : Understand shipping preferences and their timing

SELECT

Ship_Mode,

COUNT(DISTINCT Order_ID) AS Total_Orders,

ROUND(SUM(Sales), 2) AS Total_Sales,

ROUND(AVG(Sales), 2) AS Avg_Sale,

AVG(DATEDIFF(DAY, Order_Date, Ship_Date)) AS Avg_Ship_Days,

MIN(DATEDIFF(DAY, Order_Date, Ship_Date)) AS Min_Ship_Days,

MAX(DATEDIFF(DAY, Order_Date, Ship_Date)) AS Max_Ship_Days

FROM dbo.Superstore

GROUP BY Ship_Mode

ORDER BY Total_Orders DESC;

Results

Messages

	Ship_Mode	Total_Orders	Total_Sales	Avg_Sale	Avg_Ship_Days	Min_Ship_Days	Max_Ship_Days
1	Standard Class	2994	1358215.74	227.58	5	3	7
2	Second Class	964	459193.57	236.09	3	1	5
3	First Class	787	351428.42	228.5	2	1	4
4	Same Day	264	128363.12	236.4	0	0	1

OBSERVATION:

Standard Class → 60%+ of all orders, ships in 4–5 days average

Second Class → ~20% of orders, ships in 3–4 days

First Class → ~15% of orders, ships in 2 days

Same Day → <5% of orders, ships in 0 days

Premium shipping modes are used less often but carry higher-value orders.

-- **C11. State-Level Performance (Top 15 States)**

-- PURPOSE : Geographic performance deep-dive

SELECT TOP 15

State,

Region,

COUNT(DISTINCT Order_ID) AS Total_Orders,

ROUND(SUM(Sales), 2) AS Total_Sales,

ROUND(SUM(Profit), 2) AS Total_Profit,

ROUND(SUM(Profit) * 100.0

/ NULLIF(SUM(Sales),0), 2) AS Profit_Margin_Pct

FROM dbo.Superstore

GROUP BY State, Region

ORDER BY Total_Sales DESC;

Results		Messages				
	State	Region	Total_Orders	Total_Sales	Total_Profit	Profit_Margin_Pct
1	California	West	1021	457687.63	76381.39	16.69
2	New York	East	562	310876.27	74038.55	23.82
3	Texas	Central	487	170188.05	-25729.36	-15.12
4	Washington	West	256	138641.27	33402.65	24.09
5	Pennsylvania	East	288	116511.91	-15559.96	-13.35
6	Florida	South	200	89473.71	-2979.3	-3.33
7	Illinois	Central	276	80166.1	-12607.89	-15.73
8	Ohio	East	236	78258.14	-16971.38	-21.69
9	Michigan	Central	117	76269.61	24463.19	32.07
10	Virginia	South	115	70636.72	18597.95	26.33
11	North Carolina	South	136	55603.16	-7490.91	-13.47
12	Indiana	Central	73	53555.36	18382.94	34.33
13	Georgia	South	91	49095.84	16250.04	33.1
14	Kentucky	South	61	36591.75	11199.7	30.61
15	New Jersey	East	61	35764.31	9772.91	27.33

OBSERVATION:

California → Highest sales state (~\$457K)

New York → Second highest

Texas → High sales but NEGATIVE total profit (-\$25K)

Ohio → Also shows loss despite decent sales volume

Texas and Ohio are key states to investigate for margin improvement.

-- C12. Average Discount by Sub-Category

-- PURPOSE : Identify where the company over-discounts

SELECT

Sub_Category,

ROUND(AVG(Discount), 4) AS Avg_Discount,

ROUND(SUM(Sales), 2) AS Total_Sales,

ROUND(SUM(Profit), 2) AS Total_Profit,

COUNT(*) AS Transactions

FROM dbo.Superstore

GROUP BY Sub_Category

ORDER BY Avg_Discount DESC;

Results		Messages			
	Sub_Category	Avg_Discount	Total_Sales	Total_Profit	Transactions
1	Binders	0.3723	203412.73	30221.76	1523
2	Machines	0.3061	189238.63	3384.76	115
3	Tables	0.2613	206965.53	-17305.48	319
4	Bookcases	0.2111	114880	-3472.56	228
5	Chairs	0.1702	328449.1	26590.17	617
6	Appliances	0.1665	107532.16	18138.01	466
7	Copiers	0.1618	149528.03	55617.82	68
8	Phones	0.1546	330007.05	44515.73	889
9	Furnishings	0.1383	91705.16	13059.14	957
10	Fasteners	0.082	3024.28	949.52	217
11	Envelopes	0.0803	16476.4	6964.18	254
12	Accessories	0.0785	167380.32	41936.64	775
13	Supplies	0.0768	46673.54	-1189.1	190
14	Art	0.0749	27118.79	6527.79	796
15	Paper	0.0749	78479.21	34053.57	1370
16	Storage	0.0747	223843.61	21278.83	846
17	Labels	0.0687	12486.31	5546.25	364

OBSERVATION:

Binders, Tables, and Bookcases have the highest average discounts.

All three are associated with losses or thin profit margins.

INSIGHT: There is a direct correlation between high discount rates and negative profitability in this dataset.